# The Standard Model Gauge Group from Vortex Knot Topology

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**Abstract**

We propose that the gauge group SU(3) × SU(2) × U(1) of the Standard Model can be identified with the crossing exchange algebra of the two simplest vortex topologies in a superfluid vacuum: the trefoil knot and the Hopf link. We show that continuous strand exchange at the three self-crossings of the trefoil generates the Lie algebra $\mathfrak{su}(3)$, with the Gell-Mann matrices recovered as crossing transpositions and their commutators. The two inter-crossings of the Hopf link, lying in perpendicular planes in the symmetric embedding, generate $\mathfrak{su}(2)$, while the linking phase provides $\mathfrak{u}(1)$. We argue that colour confinement corresponds to the topological stability of the trefoil, and that quark degrees of freedom are associated with the three crossings. As consequences of this gauge structure, we obtain the fine structure constant $\alpha = 1/(\sqrt{2}\,\kappa^2)$ (where $\kappa = \pi^2$ is the torus aspect ratio, 0.52% from experiment), the Weinberg angle $\sin^2\theta_W = 0.222$ (4%), the W/Z mass ratio to 0.07%, and the Cabibbo angle $\sin^2\theta_C = 7\alpha$ (0.05%). The crossing number of the carrier determines the gauge rank, predicting a tower SU($N$) at higher $N$: SU(4) (Pati–Salam) for $N = 4$ and SU(5) (Georgi–Glashow) for $N = 5$. Only $N = 2$ and $N = 3$ are topologically stable at low energy, selecting the Standard Model as the minimal gauge structure compatible with the simplest vortex defects. The Gross–Pitaevskii nonlinearity at the Hopf link crossings provides the Higgs potential, with the condensate density playing the role of the Higgs field; electroweak symmetry breaking is the suppression of one crossing channel as the condensate stiffens at low temperature.

## 1 Introduction

The gauge group $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ is the foundation of the Standard Model, yet its origin remains unexplained. Why these groups? Why this product structure? Why not SU(5) or SO(10)?

We propose a topological answer. In the Null Worldtube (NWT) framework, elementary particles are quantised vortex knots in a superfluid vacuum described by the Gross–Pitaevskii energy functional [1, 2, 3]. Baryons reside on trefoil knot carriers ($n_q = 3$ crossings), mesons on Hopf link carriers ($n_q = 2$ crossings). The gauge interactions arise from topology-changing operations at these crossings.

The central result of this paper is:

**Theorem 1** (Gauge algebra from crossing exchanges). *Let $K$ be a carrier whose minimal-crossing diagram has $N$ crossings in which every pair of crossings shares at least one strand (so that the transpositions generate $S_N$). Assign to each crossing a continuous strand-exchange operator $U_{ij}(t) = \exp(itT_{ij})$ acting on the $N$-dimensional space of crossing labels. Then the generators $T_{ij}$, together with their nested commutators, span:*

- $\mathfrak{su}(N)$ *for a knot (self-crossings);*
- $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ *for a link with 2 inter-crossings in perpendicular planes.*

*For the trefoil ($N = 3$) and the Hopf link ($N = 2$):*

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1) = \textit{Lie algebra of } \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y. \tag{1}$$

The remainder of this paper proves this theorem, derives the coupling constants as consequences of the crossing geometry, and identifies the predictions that distinguish this proposal from other approaches.

# 2 Dynamical Framework

The NWT vacuum is a relativistic superfluid with order parameter $\psi$ governed by the Gross–Pitaevskii energy functional

$$E[\psi] = \int \left[ \frac{\hbar^2}{2m^*} |\nabla\psi|^2 + \frac{g}{2} \left(|\psi|^2 - n_0\right)^2 \right] d^3x, \tag{2}$$

where $m^* = m_e/\sqrt{2}$ is the condensate effective mass (the electron mass splits equally between the amplitude and phase degrees of freedom of the complex field: $m_e^2 = 2m^{*2}$). The vacuum is Lorentz-invariant ($c_s = c$). Vortex lines—one-dimensional zeros of $|\psi|$—are topologically protected: they cannot end in the bulk and their knot type is preserved by the dynamics.

Particles are stable vortex configurations on a torus of major radius $R$ and tube radius $r = R/\kappa$, with $\kappa = \pi^2$ [1]. The carrier hierarchy—unknot, Hopf link, trefoil, figure-eight, $5_1$—determines the particle sector (lepton, meson, baryon, tetraquark, pentaquark) [4].

At a carrier *crossing*, two strands of the vortex line approach within a tube diameter. The three elementary topology changes at a crossing are the Reidemeister moves: twist (R1), strand exchange (R2), and strand slide (R3). We focus on R2 (strand exchange), which is the mechanism for gauge interactions.

# 3 $\mathrm{SU}(3)$ from the Trefoil

## 3.1 Crossing generators

The trefoil $T(2, 3)$ has three crossings, labelled 0, 1, 2, equally spaced by the three-fold symmetry of the knot. At each crossing, two strands of the *same* vortex line pass each other.

A continuous strand exchange at crossing $k$ between strands $i$ and $j$ is parameterised by an amplitude $t \in [0, \pi/2]$:

$$U_{ij}(t) = \exp(i\, t\, T_{ij}), \tag{3}$$

where $T_{ij}$ is the generator of the exchange. At $t = 0$ the strands pass unperturbed; at $t = \pi/2$ they fully reconnect.

In the three-dimensional space spanned by the crossing labels $\{|0\rangle, |1\rangle, |2\rangle\}$, the exchange generators are:

$$T_{01} = \frac{\lambda_1}{2}, \qquad T_{02} = \frac{\lambda_4}{2}, \qquad T_{12} = \frac{\lambda_6}{2}, \tag{4}$$

where $\lambda_1$, $\lambda_4$, $\lambda_6$ are the real off-diagonal Gell-Mann matrices.

## 3.2 Generation of the full algebra

**Proposition 1.** *The three generators $T_{01}$, $T_{02}$, $T_{12}$ generate the complete Lie algebra $\mathfrak{su}(3)$ via nested commutators.*

*Proof.* Level 0 (crossing exchanges): $\lambda_1$, $\lambda_4$, $\lambda_6$.

Level 1 (commutators of Level-0 generators):

$$\begin{aligned}
[\lambda_1, \lambda_4] &= i\lambda_7, \\
[\lambda_1, \lambda_6] &= -i\lambda_5, \\
[\lambda_4, \lambda_6] &= i\lambda_2.
\end{aligned} \tag{5}$$

This produces $\lambda_2$, $\lambda_5$, $\lambda_7$ (the imaginary off-diagonal generators).

Level 2 (commutators of Level-0 with Level-1):

$$\begin{aligned}
[\lambda_1, \lambda_2] &= 2i\lambda_3, \\
[\lambda_4, \lambda_5] &= i\lambda_3 + i\sqrt{3}\,\lambda_8.
\end{aligned} \tag{6}$$

This produces $\lambda_3$ and $\lambda_8$ (the Cartan subalgebra / diagonal generators).

Total: all eight Gell-Mann matrices are generated. Since these form a basis for $\mathfrak{su}(3)$, the crossing exchange generators span the complete algebra. □

This is a specific instance of the standard result that the real off-diagonal generators of $\mathfrak{su}(N)$ generate the full algebra for any $N \geq 2$. The physical content of Theorem 1 is the identification of these generators with strand exchanges at trefoil crossings.

### 3.3 Physical interpretation

The three crossings of the trefoil are associated with the three colour degrees of freedom:

$$|0\rangle \equiv |r\rangle, \qquad |1\rangle \equiv |g\rangle, \qquad |2\rangle \equiv |b\rangle. \tag{7}$$

The eight gluons correspond to the eight independent exchange modes (three real transpositions, three phase-shifted transpositions, two diagonal differences).

Colour confinement has a direct topological explanation: the trefoil is the simplest non-trivial knot and cannot be simplified by any ambient isotopy. A baryon's three "quarks" (the three crossings) are permanently linked by the knot topology. To separate them would require cutting the vortex line—an operation forbidden by the GPE dynamics (vortex lines cannot end in the bulk).

The $\mathbb{Z}_3$ centre of SU(3)—the discrete symmetry that distinguishes quarks from antiquarks—is the three-fold rotational symmetry of the trefoil.

## 4 $\mathrm{SU}(2) \times \mathrm{U}(1)$ from the Hopf Link

### 4.1 Perpendicular crossings

The Hopf link consists of two unknotted rings, linked once. It has two inter-crossings—points where ring A passes through ring B. In the symmetric embedding, these two crossings lie in *perpendicular planes*—a natural geometric realisation of the linking, in which ring A passes through the interior of ring B from two orthogonal directions.

In the two-dimensional space $\{|A\rangle, |B\rangle\}$ of ring labels:

$$\begin{aligned}
T_1 &= \frac{\sigma_1}{2} \qquad \text{(crossing 1, exchange in the } xz\text{-plane)}, \\
T_2 &= \frac{\sigma_2}{2} \qquad \text{(crossing 2, exchange in the } yz\text{-plane)},
\end{aligned} \tag{8}$$

where $\sigma_1$, $\sigma_2$ are Pauli matrices. The perpendicularity of the crossings forces $T_2 = \sigma_2/2$ rather than a second copy of $\sigma_1/2$.

### 4.2 Generation of $\mathfrak{su}(2)$

The commutator

$$[T_1, T_2] = \left[\frac{\sigma_1}{2}, \frac{\sigma_2}{2}\right] = \frac{i\sigma_3}{2} \equiv iT_3 \tag{9}$$

produces the third generator $T_3 = \sigma_3/2$. The three generators satisfy $[T_i, T_j] = i\epsilon_{ijk} T_k$, the standard $\mathfrak{su}(2)$ commutation relations.

### 4.3 The linking phase and U(1)

The Hopf link has linking number lk = 1. The overall phase associated with one ring winding through the other defines a U(1) generator

$$Q = \frac{\mathrm{lk}}{2} I_2 \tag{10}$$

that commutes with all SU(2) generators: $[Q, T_i] = 0$. The full algebra is therefore $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

We note that this U(1) is a *topological* U(1) (winding number), not yet equipped with the hypercharge assignments of the Standard Model. The specific charge spectrum $Y = -1, -1/3, 2/3, \ldots$ must come from the interaction of the linking phase with the mode quantum numbers—a calculation we defer to future work. What is established here is the *algebraic structure* $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$, not the full charge assignments.

The physical $Z^0$ and photon $\gamma$ are mixtures of the SU(2) neutral generator $T_3$ and the hypercharge $Q$:

$$Z = \cos\theta_W T_3 - \sin\theta_W Q, \qquad \gamma = \sin\theta_W T_3 + \cos\theta_W Q. \tag{11}$$

## 5 The Combined Gauge Group

Since baryons reside on trefoil carriers and mesons/electroweak bosons on Hopf link carriers, the full gauge structure is the direct sum:

$$\boxed{\mathfrak{su}(3)_C \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y = \text{crossing algebra of trefoil} \oplus \text{crossing algebra of Hopf link.}} \tag{12}$$

The distinction between the two factors reflects the *topology* of the carrier:

| Carrier | Crossing type | Algebra | Gauge group |
|---|---|---|---|
| Trefoil (knot) | Self-crossings | $\mathfrak{su}(3)$ | $\mathrm{SU}(3)_C$ |
| Hopf link (link) | Inter-crossings | $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ | $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ |

The product structure $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ emerges as a minimal gauge structure compatible with the two simplest non-trivial vortex topologies—the trefoil (simplest knot) and the Hopf link (simplest link with $|\mathrm{lk}| > 0$).

## 6 Electroweak Symmetry Breaking

The gauge group $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ is broken to $\mathrm{U}(1)_{\mathrm{EM}}$ at low energy. In the Standard Model, this requires an additional scalar field (the Higgs doublet) with a Mexican-hat potential. In NWT, no additional independent field is introduced: the role of the Higgs is played by the GPE energy functional (2), whose nonlinearity supplies the Mexican-hat potential.

### 6.1 The crossing condensate as the Higgs field

At each Hopf link crossing, the condensate field $\psi$ has a specific density $|\psi|^2$ in the region between the two strands. We identify this crossing condensate with the Higgs field:

$$\phi(x) \longleftrightarrow |\psi(x)|^2\Big|_{\text{crossing region}}. \tag{13}$$

The GPE potential restricted to the crossing region is

$$V(\psi_{\text{cross}}) = -g_{\text{NL}}\, n_0\, |\psi|^2 + \frac{g_{\text{NL}}}{2}\, |\psi|^4, \tag{14}$$

which has the form $V = -\mu^2|\phi|^2+\lambda|\phi|^4$ with $\mu^2 = g_{\text{NL}}n_0$ and $\lambda = g_{\text{NL}}/2$. This is the Mexican-hat potential.

### 6.2 Symmetry breaking from crossing suppression

The two Hopf crossings generate SU(2) when both exchange channels are active. Parameterise the exchange amplitude at crossing 2 by $\eta \in [0, 1]$:

$$T_1 = \frac{\sigma_1}{2}, \qquad T_2(\eta) = \eta\,\frac{\sigma_2}{2}, \qquad T_3(\eta) = [T_1, T_2(\eta)]/i = \eta\,\frac{\sigma_3}{2}. \tag{15}$$

At $\eta = 1$ (high temperature, crossing condensate thermally depleted): both channels active, full $\mathrm{SU}(2) \times \mathrm{U}(1)$. At $\eta = 0$ (low temperature, condensate at equilibrium density): crossing 2 suppressed by the energy barrier of the stiff condensate, only $\mathrm{U}(1)_{\text{EM}}$ survives.

The electromagnetic U(1) is protected because it corresponds to the *linking phase* $Q$ (Eq. 10), which is topological: the linking number $\mathrm{lk} = 1$ is preserved regardless of the crossing dynamics. This is why the photon is exactly massless—its gauge symmetry is not a dynamical exchange but a topological invariant.

### 6.3 Goldstone modes and the Higgs boson

The three broken generators ($T_1$, $T_2$, $T_3$ at $\eta < 1$) correspond to three Goldstone modes—phase fluctuations of $\psi$ at the crossing—which are absorbed as the longitudinal polarisations of the $\mathrm{W}^{\pm}$ and $\mathrm{Z}^0$. The remaining degree of freedom—the *amplitude* fluctuation of $|\psi|$ at the crossing—is the physical Higgs boson.

The Higgs self-coupling is

$$\lambda = \frac{m_H^2}{2v^2} = \frac{(m_H/m_W)^2\, g^2}{8} = 0.125, \tag{16}$$

using the NWT masses $m_H = 125.0$ GeV, $m_W = 80.4$ GeV, and $g = \sqrt{4\pi\alpha}/\sin\theta_W = 0.643$. The SM value is $\lambda \approx 0.13$ (3.8% agreement). We do not derive the full gauge boson mass matrix here; the identification is based on degree-of-freedom counting and the symmetry structure of the crossing condensate.

### 6.4 The electroweak phase transition

Electroweak symmetry restoration at high temperature corresponds to thermal depletion of the condensate at Hopf link crossings. When $k_B T$ exceeds the condensate binding energy at the crossing, vortex reconnection becomes easy, both exchange channels reactivate, and the full $\mathrm{SU}(2) \times \mathrm{U}(1)$ is restored. This is the electroweak phase transition of the early universe, occurring in the NWT framework as a specific instance of vortex crystallisation [5].

## 7 Consequences: Coupling Constants and Masses

As consistency checks, we note that the crossing geometry on a torus with aspect ratio $\kappa = R/r = \pi^2$ reproduces Standard Model coupling constants and mass ratios. No parameter tuning was performed beyond fixing $\kappa$; sensitivity to the functional forms is analysed in [2, 5]:

| Quantity | NWT expression | Value | Error |
|---|---|---|---|
| $\alpha$ | $1/(\sqrt{2}\,\kappa^2)$ | 1/137.76 | 0.52% |
| $\alpha_s$ | $K_0(2\beta/\kappa)/\ln(8\kappa)$ | 0.69 | $O(1)$ ✓ |
| $\sin^2\theta_W$ | $1-(m_W/m_Z)^2$ from $\Delta m = 1$ | 0.222 | 4% |
| $m_Z/m_W$ | $\beta(11)\ln(8\beta(11))/\beta(10)\ln(8\beta(10))$ | 1.1337 | 0.07% |
| $(g-2)/2$ | $\alpha/(2\pi)$ | 0.001155 | 0.52% |
| $\sin^2\theta_C$ | $7\alpha$ | 0.0508 | 0.05% |
| $\sigma_{pp}$ | $2\pi R_p^2$ | 44 mb | $\sim 10\%$ |

The fine structure constant $\alpha = 1/(\sqrt{2}\,\kappa^2)$ arises from the R1 (twist) operator on the tube surface, with $(r/R)^2$ from the solid angle, 1/2 from the $\cos\theta$ selection rule, and $\sqrt{2}$ from the condensate amplitude–phase equipartition [5].

The W, Z, and Higgs bosons are identified as high-excitation Hopf(2) meson states with poloidal mode $q_m = 5$ and $n_q^{q_{\text{eff}}} = 2^{10} = 1024$, reproducing their masses to sub-percent accuracy [5].

The Cabibbo angle satisfies $\sin^2\theta_C = n_e\alpha$, where $n_e = 7$ is the electron's effective winding number on the trefoil $T(2,3)$. This connects quark mixing (CKM, from the Hopf link's mod-2 parity) to lepton structure (electron winding) through the common carrier geometry.

## 8 The Gauge Rank Tower

The theorem generalises to carriers with $N > 3$ crossings: a knot with $N$ self-crossings generates $\mathfrak{su}(N)$. This predicts a tower of gauge groups:

| $N$ | Carrier | Gauge group | Status |
|---|---|---|---|
| 2 | Hopf link | SU(2) × U(1) | Observed (electroweak) |
| 3 | Trefoil | SU(3) | Observed (colour) |
| 4 | Figure-eight ($4_1$) | SU(4) | Pati–Salam model [11] |
| 5 | $5_1$ torus knot | SU(5) | Georgi–Glashow GUT [12] |

The $N = 4$ and $N = 5$ carriers are the figure-eight knot and the $5_1$ torus knot, which host tetraquarks and pentaquarks respectively. In the NWT framework, these carriers are either unstable or too massive to produce observable effects at current energies, explaining why SU(4) and SU(5) gauge bosons have not been detected.

This provides a topological reason for the *absence* of grand unification at accessible energies: the higher-$N$ carriers exist in principle but decay too rapidly (the linking-number stability criterion $|\text{lk}| > 0$ is harder to satisfy for complex knots [4]).

## 9 Discussion

### 9.1 Relation to other approaches

The idea of connecting gauge groups to topology has a long history. Kelvin's vortex atoms [7] first linked particle identity to knot type. Finkelstein [8] and Faddeev–Niemi [9] explored knotted solitons as particle models. Volovik [10] showed that emergent gauge fields arise from superfluid topology. The present work goes further by deriving the *specific* gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ from the crossing structure of specific knots.

### 9.2 What is derived vs. postulated

| Result | Status | Dependence |
|---|---|---|
| $\mathfrak{su}(N)$ from $N$-crossing algebra | Mathematical identity | Standard Lie theory |
| All 8 Gell-Mann matrices from 3 generators | Verified numerically | Crossing basis choice |
| $\mathfrak{su}(2)\oplus\mathfrak{u}(1)$ from Hopf | Mathematical identity | Perpendicularity |
| Coupling constants ($\alpha$, $\alpha_s$, etc.) | Geometric parameterisation | $\kappa = \pi^2$ |
| Particles are vortex knots in GPE | Physical postulate | Condensate model |
| Trefoil = baryon, Hopf = meson | Physical identification | Carrier hierarchy |
| Crossing exchanges = gauge interactions | Proposed correspondence | Central claim |

### 9.3 From crossing algebra to gauge fields

The crossing exchange algebra generates $\mathfrak{su}(N)$, but a gauge theory requires *local* gauge fields $A_\mu(x)$. We sketch the connection; a full development will appear elsewhere.

A vortex mode propagating along the carrier encounters crossings at discrete positions $\{s_k\}$ on the knot path. At each crossing, the mode state $|\psi\rangle$ is acted upon by the exchange operator $U_k = \exp(it_k T_k)$. The parallel transport of $|\psi\rangle$ from one crossing to the next is

$$|\psi(s_{k+1})\rangle = U_k\,|\psi(s_k)\rangle = \exp(it_k T_k)\,|\psi(s_k)\rangle. \tag{17}$$

This is a *lattice gauge theory* [13] on the one-dimensional knot path, with the crossings as lattice sites and $U_k$ as the link variables. The Wilson loop around the carrier is

$$W = \mathrm{tr}\left(\prod_{k=1}^{N} U_k\right) = \mathrm{tr}\left(\prod_k \exp(it_k T_k)\right), \tag{18}$$

which is gauge-invariant by construction.

In the continuum limit (many crossings, fine spacing), the discrete product becomes a path-ordered exponential $\mathcal{P}\exp(i\oint A_\mu\,dx^\mu)$, recovering the standard gauge connection. For the trefoil with 3 crossings, the lattice is maximally coarse—but the algebra is exact because the commutator structure of $\mathfrak{su}(3)$ does not depend on lattice refinement. Locality arises because the exchange operators act only when the mode wavefunction overlaps the crossing region, which is spatially localised within approximately one tube diameter $2r$ on the vortex.

**Representations.** The fundamental representation acts on the $N$-dimensional space of crossing labels. For the trefoil: quarks transform as triplets $\mathbf{3}$ because they *are* the three crossings. The exchange operators live in the adjoint representation $\mathbf{8}$, corresponding to the eight gluon degrees of freedom. Antiquarks ($\bar{\mathbf{3}}$) correspond to the conjugate crossing labels (reversed strand ordering at each crossing).

### 9.4 The algebra-to-dynamics gap

We emphasise that the present work establishes an *algebraic* correspondence, not a complete gauge theory. The missing bridge is a demonstration that the crossing exchange operators couple to matter fields in the same way as the Yang–Mills gauge bosons of the Standard Model—specifically, that conserved colour currents flow along the trefoil strands and that the covariant derivative on the vortex tube surface reproduces the minimal coupling of QCD and electroweak theory. This is the subject of ongoing work using the Level-3 GPE solver. We present the algebraic result here because the dimensional coincidence (8 generators from 3 crossings, 4 generators from 2 crossings) is too precise to be accidental and because it passes non-trivial structural tests (the correct commutation relations, the correct centre, and the correct distinction between knot and link sectors).

### 9.5 Predictions

1. No proton decay (trefoil is topologically stable).
2. $\sin^2 \theta_C = 7\alpha$ (connects two independently measured quantities, falsifiable to 0.05%).
3. Gauge rank tower: if tetraquark states exhibit SU(4)-like selection rules, this would support the framework.
4. Proton form factor zero-crossing from trefoil diffraction.
5. Carriers with $|\mathrm{lk}| = 0$ (Whitehead link, Borromean rings) produce no stable particles.

## 10 Conclusion

The Standard Model gauge group is not a free input. It is the crossing exchange algebra of the two simplest topological defects in a superfluid vacuum—the trefoil knot and the Hopf link. Three self-crossings give SU(3). Two perpendicular inter-crossings give SU(2). The linking phase gives U(1). The Gross–Pitaevskii nonlinearity at the Hopf link crossings provides the Higgs potential, breaking $\mathrm{SU}(2) \times \mathrm{U}(1)$ to $\mathrm{U}(1)_{\mathrm{EM}}$ with $\lambda = 0.125$ (3.8% from experiment).

Nature uses the simplest non-trivial knot and the simplest non-trivial link because these are the lowest-energy topological defects in the condensate. If the identification of strand exchanges with gauge interactions is correct, $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is not a choice—it is a consequence of using the simplest carriers. Its breaking pattern is not imposed—it is the dynamics of the condensate at the crossings. Whether this topological origin can be promoted to a complete gauge theory remains the central open question.

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## Data and Code Availability

All analysis code is available at `https://github.com/JimGalasyn/null-worldtube`.